

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

**Duration : 1 Hour
Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I Thamaraiselvi S (192511042) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Thamaraiselvi S

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

1. Constraint-Based Problem Solving using Backtracking Search

Aim

To assign three doctors (D1, D2, D3) to three shifts (Morning, Afternoon, Night) while satisfying all the given constraints using the Backtracking Search algorithm.

Variables

Doctors = {D1, D2, D3}

Shifts = {Morning, Afternoon, Night}

Constraints

- D1 cannot work Night shift.
- D2 must work before D3.
- Only one doctor can work in one shift.
- D3 cannot work Morning shift.
- Every doctor should be assigned exactly one shift.

Algorithm (Backtracking)

1. Start with an empty assignment.
2. Assign a shift to the first doctor.
3. Check whether all constraints are satisfied.
4. If a constraint is violated, backtrack and try another shift.
5. Continue until every doctor gets one valid shift.
6. Stop when all constraints are satisfied.

Step-by-Step Execution

Step 1:

Assign

D1 → Morning

Constraint Check

- D1 is not Night ✓
- Shift available ✓

Assignment

Doctor Shift

D1 Morning

Step 2:

Assign

D2 → Afternoon

Constraint Check

- Shift unique ✓
- D2 before D3 (temporary) ✓

Assignment

Doctor	Shift
D1	Morning
D2	Afternoon

Step 3:

Assign

D3 → Night

Constraint Check

- D3 not Morning ✓
- D2 before D3 ✓
- Shift unique ✓

Assignment

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

Constraint Satisfaction Table

Constraint	Result
D1 ≠ Night	Satisfied
D2 before D3	Satisfied
One doctor per shift	Satisfied
D3 ≠ Morning	Satisfied
All doctors assigned	Satisfied

Advantages of Backtracking

- Finds a valid solution efficiently.
- Eliminates invalid assignments early.
- Saves time by avoiding unnecessary searches.
- Suitable for scheduling and timetable problems.

Final Answer

Doctor	Assigned Shift
D1	Morning
D2	Afternoon
D3	Night

2. Robot Navigation using BFS Search

Aim

To find the shortest path from Start (S) to Goal (G) using Breadth First Search (BFS).

Grid

```
S . . X .  
. X . X .  
. . . . .  
X X . X .  
. . . . G
```

Constraints:

- Move only Up, Down, Left and Right.
- Cannot move through obstacles.
- Every move costs 1.
- Manhattan Distance heuristic:

Formula:

Manhattan Distance = $|x_1 - x_2| + |y_1 - y_2|$

Start = (0,0)

Goal = (4,4)

Distance = $|4 - 0| + |4 - 0| = 8$

BFS Algorithm:

1. Put the Start node into the queue.
2. Remove the first node.
3. Visit all unvisited neighbouring nodes.
4. Add neighbours to the queue.
5. Repeat until Goal is found.

Node Expansion:

Step	Expanded Node	Queue
1	(0,0)	(0,1),(1,0)
2	(0,1)	(1,0),(0,2)
3	(1,0)	(0,2),(2,0)
4	(0,2)	(2,0),(1,2)
5	(2,0)	(1,2),(2,1)
6	(1,2)	(2,1),(2,2)
7	(2,1)	(2,2),(2,3)
8	(2,2)	(2,3),(3,2)

Step	Expanded Node	Queue
9	(2,3)	(3,2),(2,4)
10	(3,2)	(2,4),(4,2)
11	(4,2)	(4,3)
12	(4,3)	(4,4)
13	(4,4)	Goal Found

Optimal Path:

(0,0)

↓

(1,0)

↓

(2,0)

→

(2,1)

→

(2,2)

↓

(3,2)

↓

(4,2)

→

(4,3)

→

(4,4)

Cost Calculation:

Number of Moves = 8

Cost per Move = 1

Total Cost = **8**

Advantages of BFS:

- Finds the shortest path in an unweighted graph.
- Simple and complete algorithm.
- Guarantees an optimal solution when all moves have equal cost.

Final Answer:

Optimal Path: $S \rightarrow (1,0) \rightarrow (2,0) \rightarrow (2,1) \rightarrow (2,2) \rightarrow (3,2) \rightarrow (4,2) \rightarrow (4,3) \rightarrow G$

Total Cost = 8

3. Autonomous Rescue Robot

(a) Constraint Analysis

Constraint 1:

Movement only in four directions.

Effect: Reduces the number of possible actions and simplifies navigation.

Constraint 2:

Blocked cells cannot be crossed.

Effect: The robot must search for an alternative route.

Constraint 3:

Limited sensing capability.

Effect: The robot knows only neighbouring rooms and updates its map while moving.

Constraint 4:

Each movement costs 1.

Effect: The robot tries to minimize the total number of movements.

Constraint 5:

Risky zones have an additional cost of 2.

Effect: The robot prefers safer paths, even if they are slightly longer.

Constraint 6:

Dynamic knowledge update.

Effect: The robot replans its path whenever it discovers new obstacles or risks.

(b) Solution Approach:

Search Strategy

Uniform Cost Search (UCS) with Online Search Agent

Working Steps:

1. Start from the initial position (S).
2. Observe adjacent rooms.
3. Ignore blocked cells.
4. Calculate cumulative path cost.
5. Expand the node with the minimum total cost.
6. Update the environment map when new information is detected.
7. Continue until the survivor (G) is reached.

Cost Example:

Cell	Cost
Normal Cell	1
Risk Zone	3 (1+2)

The robot always selects the path with the **lowest cumulative cost**.

(c) AI Concepts

Intelligent Agent:

- Senses the environment.
- Processes available information.
- Chooses the best action.
- Acts autonomously to rescue survivors.

Rationality:

A rational agent selects the action that maximizes performance by minimizing travel cost and avoiding dangerous areas while reaching the goal.

Environment Type:

Property	Description
Observability	Partially Observable
Determinism	Partially Deterministic
Dynamics	Dynamic
State Space	Discrete
Agent Type	Single Agent

(d) Justification

Uniform Cost Search is the most suitable search algorithm because:

- It always finds the minimum-cost path.
- It considers both movement cost and additional risk cost.
- It dynamically adapts to newly discovered information.
- It is effective in partially observable environments.
- It ensures safe, intelligent, and optimal navigation.

Conclusion:

The rescue robot acts as an **intelligent and rational agent** by sensing the environment, updating its knowledge dynamically, avoiding obstacles and risky zones, and using **Uniform Cost Search** to reach the survivor with the minimum possible cost. This approach guarantees efficient, safe, and optimal rescue operations under the given constraints.